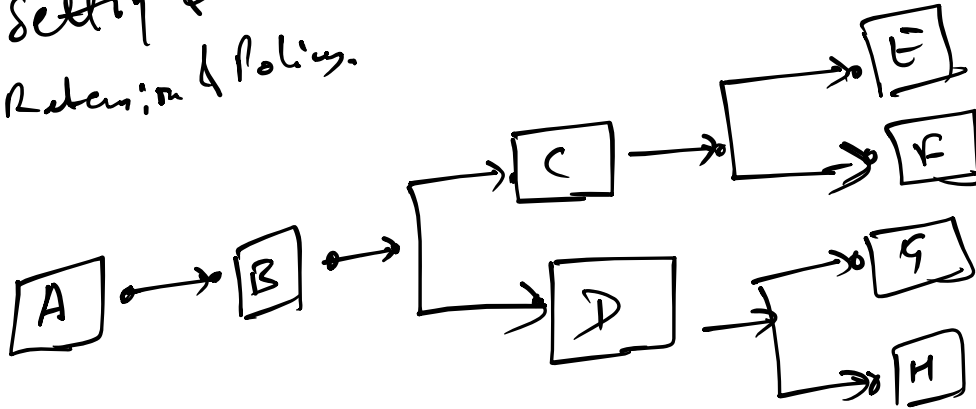


- ① Smartscape Topology.
- ② Alert & Problem Detection
- ③ Integration.
- ④ User Mng. & Access Control
- ⑤ Env. Setting & Config.
- ⑥ Data Retention & Policy.
- ⑦ Setting.
- ⑧ Workflow Action.



Provider :- Categorize external resources.

CDN
Analytics
Ads/trackers

Payment Gateway

Fonts, Image, JS Libraries

First Party → your app
3rd Party → external service.

Bucket → where DT store the data.

⑤ Davis AI Problem.

- ① Logs.
- ② Metrics
- ③ Events
- ④ User Session

Retention Rule

- ① What data is stored
- ② How long it is stored.
- ③ Where it is stored.

Why?

- ① Cost Control.
- ② Performance.
- ③ Compliance

Recommend

- ① Log → 15-30 day.
- ② Log → 7-15 d
- ③ Mem → 6-15 months
- ④ Event → 30-60 d
- ⑤ B.E → Based on use case

① Violating Sample → How many measurement points must break the threshold b/f DT raises an alert.

② Sliding Window →

③ Deactivating Samples → How many sample must return b/f DT close the alert

④ Deactivating Window →

Sliding window → 30 min → every 1 min → one sample.

30 min → 30 sample.

VS + 18 samples → Alert trigger

V.S. > 18

GC (garbage collection):

Frees memory by removing object that are no longer used.

- High GC $\rightarrow$ JVM spending too much time to do $\rightarrow$ Instead of doing activity

High GC $\uparrow$ Slow Application $\downarrow$ Coverage $\uparrow$ RT $\uparrow$

Java out of Memory (OOM): - JVM can't allocate memory anymore.

Heap full, failed $\rightarrow$ throw error.

Type of OOM:-

① Heap OOM

② Metaspace OOM

③ Direct Memory OOM.

① Heap OOM:- Too many obj.
Memory leak.

② Metaspace OOM:- Too many classes loaded.

③ Direct Memory OOM:- off-heap Memory full.

Why critical?

① Traffic Increases

② Object created faster than GC clean.

Heap fill up

(2)

(3) Heap fill up

Java Out of Threads:- Can't create new thread anymore.
↳ Each req. needs a thread.

Why?

- (1) Too many thread created
- (2) Thread Leak.
- (3) Blocking Call.

Real Impact:-

(1) App. Hung.

(2) Req. Stop.

(3) sly looks "alive" but does nothing

High traffic → more object created

GC Incremental → tries to clean

Memory still insufficient → OOM

Thread will wait → Blocked

Thread pool exhausted → Out of Thread.

Easy Travel?

Traffic spike — Users Increase.
More object created in JVM
Heap memory usage increases.

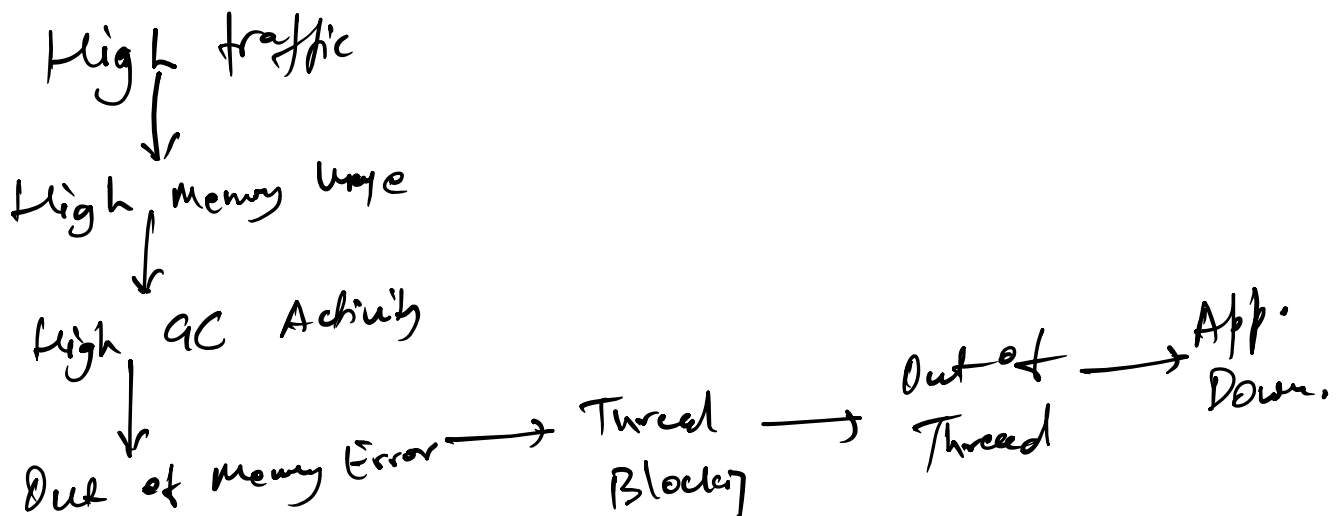
High GC Activity — JVM tries to clean memory
GC time % ↑ RT ↑
CPU ↑

Memory not enough → OOM
Service failure rate ↑
APP become unstable

Thread get Blocked → Req. waiting.
Thread not Released.

Thread pool fill up.

Out of Thread RT → infinite / time out.
APP stops responding completely.



How to fix:-

High GC

- ① Increase Heap size.
- ② Use Better GC
- ③ Optimize object creation

OOM:-

- ① Memory leak
- ② Large obj.
- ③ Wrong heap size

Out of Thread

- ① Use thread pools
- ② Tune:-
 - ① max thread.
 - ② Queue size.

Prevent chain failure

Alert eng:-

Heap usage $\geq 80\%$
GC Time $\geq 10-15\%$
Thread usage $\geq 70\%$

Don't wait for OOM.

Contains $\rightarrow$ * _ * $\rightarrow$ \$ contains (svc)
eg $\rightarrow$ svc $\rightarrow$ \$ eg (svc)
prefix $\rightarrow$ svc $\rightarrow$ \$ prefix (svc)
suffix $\rightarrow$ \$ suffix (svc.py)
 $\swarrow$
 loyalty svc.py
 $\swarrow$
other words $\leftarrow$ suffix